Hugo LAM

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French nationality - Paris, France

Objective: Seeking a **5 to 6-month internship in Mechanical Engineering** starting March 2027 as part of my Master of Science in Aerodynamics

Target Areas: External & Internal Aerodynamics, CFD, Aeroacoustics, Flow Optimization

Education

Master of Science (M2) in Engineering - Aerodynamics 2026 - 2027
Sorbonne Université - École Nationale Supérieure des Arts et Métiers (ENSAM), Paris

Relevant courses: Turbulence Dynamics and Modelling, Numerical Simulation Practice (OpenFOAM), Numerical Methods for Compressible Flows, Fundamental Aerodynamics, Aeroacoustics, Aeroelasticity

Master of Science (M1) in Engineering - Computational Mechanics 2025 - 2026
Sorbonne Université, Paris

Relevant courses: Hydrodynamics, Finite Element Analysis, Compressible Flows, Numerical Methods in Fluid Mechanics

Bachelor's degree - Major in Mechanics, Minor in AI & Data Science 2022 - 2025
Sorbonne Université, Paris | ***Honors: Assez Bien***

Relevant courses: Continuum Mechanics, Mathematical and Numerical Methods

Baccalauréat Général 2019 - 2022
Lycée Pierre-Gilles de Gennes (ENCPB), Paris | ***Honors: Bien***
Majors: Mathematics and Engineering Sciences

Work Experience

Research Intern May 2026 - July 2026
Jean le Rond d'Alembert Institute - Paris

- Numerical modelling of Rayleigh-Bénard bioconvection instabilities driven by gravity and gyrotaxis coupling, extended to phototaxis under light illumination; analysed plume dynamics (spacing, velocities, vorticity) and geometric confinement effects
- Supervised by José-Maria Fullana and Arnaud Antkowiak (based on Anaëlle Givaudan's doctoral research)

Aircraft Maintenance Intern July 2024
Cathay Pacific - Paris-Charles de Gaulle Airport

- Conducted arrival and transit walkaround inspections on Boeing 777-300ER and Boeing 747-400 aircraft, strictly adhering to manufacturer AMM (Aircraft Maintenance Manual) procedures
- Supported line maintenance operations: engine oil replenishment, exterior lighting checks and cabin safety inspections

Multipurpose Team Member (Part-time - Weekends) July 2024 (1 month), July 2025 (1 month)
Taiyaki Paris - Paris

- Handled customer service, counter sales, order processing and store organisation

Academic Projects

CFD Study of a NACA 0012 Airfoil: Aerodynamic flow simulation around a NACA 0012 airfoil across subsonic and transonic Mach numbers; post-processing of aerodynamic coefficients (C_L, C_D) and benchmark comparison against wind-tunnel data

Simulation of Bacterial Pattern Formation: Numerical solution in Python of reaction–diffusion models (Gray–Scott, Kawasaki–Murray) using finite differences and an explicit Euler scheme; study of how diffusion and reaction parameters drive pattern formation (Turing instability)

Skills & Languages

- **Languages:** French (Native), English (B2)
- **Programming Languages:** Python, C
- **Simulation & Meshing:** OpenFOAM, Basilisk, Gmsh
- **Data Analysis & Visualisation:** ParaView, Jupyter Notebook, Spyder